Homework 8

1. Convergence of Gradient Descent for Ridge Regression

Let $A \in \mathbb{R}^{m \times n}$, $\vec{y} \in \mathbb{R}^m$, and $\lambda > 0$. Consider a slight variation of the ridge regression problem where the least squares loss is normalized by the number of data points:

$$\min_{\vec{x} \in \mathbb{R}^n} f_\lambda(\vec{x}) \qquad \text{where} \qquad f_\lambda(\vec{x}) \doteq \frac{1}{2}\left\{\frac{1}{m}\|A\vec{x} - \vec{y}\|_2^2 + \lambda \|\vec{x}\|_2^2\right\}. \tag{1}$$

In this problem, we will examine the behavior of gradient descent (GD) on this problem, and in particular the interplay between the learning rate $\eta > 0$ and regularization parameter $\lambda > 0$ in determining convergence.

(a) Show that the unique solution to the problem in Equation (1) is

$$\vec{x}_\lambda^\star = \left(A^\top A + \lambda m I\right)^{-1} A^\top \vec{y}. \tag{2}$$

Solution: The function $\lambda \|\vec{x}\|_2^2$ in Equation (1) is strictly convex, so f_λ is strictly convex. Thus the problem in Equation (1) has strictly convex objective and convex feasible set $\mathbb{R}^n$, so it has at most one solution. And we can find a solution by setting the gradient to $\vec{0}$:

$$\begin{aligned}
\nabla f_\lambda(\vec{x}_\lambda^\star) &= \frac{1}{m}A^\top(A\vec{x}_\lambda^\star - \vec{y}) + \lambda \vec{x}_\lambda^\star && (3)\\
&= \left(\frac{A^\top A}{m} + \lambda I\right)\vec{x}_\lambda^\star - \frac{1}{m}A^\top \vec{y} && (4)\\
\implies \left(\frac{A^\top A}{m} + \lambda I\right)\vec{x}_\lambda^\star &= \frac{1}{m}A^\top \vec{y} && (5)\\
\implies \left(A^\top A + \lambda m I\right)\vec{x}_\lambda^\star &= A^\top \vec{y} && (6)\\
\implies \vec{x}_\lambda^\star &= \left(A^\top A + \lambda m I\right)^{-1} A^\top \vec{y}. && (7)
\end{aligned}$$

(b) Show that the GD update

$$\vec{x}_{t+1} = \vec{x}_t - \eta\left(\frac{1}{m}A^\top(A\vec{x}_t - \vec{y}) + \lambda \vec{x}_t\right) \tag{8}$$

can be rearranged into the form

$$\vec{x}_{t+1} - \vec{x}_\lambda^\star = \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)(\vec{x}_t - \vec{x}_\lambda^\star). \tag{9}$$

Use this to show that

$$\vec{x}_t - \vec{x}_\lambda^\star = \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star). \tag{10}$$

for every positive integer t.

Solution: We have

$$\vec{x}_{t+1} = \vec{x}_t - \eta\left(\frac{1}{m}A^\top(A\vec{x}_t - \vec{y}) + \lambda \vec{x}_t\right) \tag{11}$$

$$= \vec{x}_t - \eta \cdot \frac{A^\top A}{m}\vec{x}_t + \eta \cdot \frac{1}{m}A^\top \vec{y} + \eta\lambda\vec{x}_t \tag{12}$$

$$= \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)\vec{x}_t + \eta \cdot \frac{1}{m}A^\top \vec{y} \tag{13}$$

$$\implies \vec{x}_{t+1} - \vec{x}_\lambda^\star = \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)\vec{x}_t + \eta \cdot \left(\frac{A^\top A}{m} + \lambda I\right)\vec{x}_\lambda^\star - \vec{x}_\lambda^\star \tag{14}$$

$$= \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)(\vec{x}_t - \vec{x}_\lambda^\star). \tag{15}$$

Iterating this relation obtains the second equality.

(c) We now discuss the insight that the SVD can give us regarding the convergence of GD. Let $A = U\Sigma V^\top$ be a full SVD of A. Let $\vec{z}_t = V^\top \vec{x}_t$ and $\vec{z}_\lambda^\star = V^\top \vec{x}_\lambda^\star$. Show that

$$\vec{z}_t - \vec{z}_\lambda^\star = \left(I - \eta\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)\right)^t (\vec{z}_0 - \vec{z}_\lambda^\star), \tag{16}$$

and, moreover, show that for each $i \in \{1, \ldots, n\}$, we have

$$(\vec{z}_t)_i - (\vec{z}_\lambda^\star)_i = \left(1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right)^t ((\vec{z}_0)_i - (\vec{z}_\lambda^\star)_i) \tag{17}$$

where $\sigma_i\{A\}$ is the ith largest singular value of A. This shows that the *rate of convergence* of $\vec{z}_t$ to $\vec{z}_\lambda^\star$ along the ith component is influenced by the interaction between $\sigma_i\{A\}$, λ, and η, but critically no other singular values. Thus, one considers the V basis to be the "natural" basis for thinking about GD for ridge regression.

Solution: If $A = U\Sigma V^\top$ then

$$A^\top A = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top\Sigma V^\top. \tag{18}$$

Thus we have

$$\vec{x}_t - \vec{x}_\lambda^\star = \left(I - \eta\left(\frac{A^\top A}{m} + \lambda I\right)\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star) \tag{19}$$

$$= \left(I - \eta\left(\frac{V\Sigma^\top\Sigma V^\top}{m} + \lambda I\right)\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star) \tag{20}$$

$$= \left(I - \eta\left(V\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)V^\top\right)\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star) \tag{21}$$

$$= \left(I - \eta V\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)V^\top\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star) \tag{22}$$

$$= \left(V\left(I - \eta\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)\right)V^\top\right)^t (\vec{x}_0 - \vec{x}_\lambda^\star) \tag{23}$$

$$= V\left(I - \eta\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)\right)^t V^\top(\vec{x}_0 - \vec{x}_\lambda^\star) \tag{24}$$

$$\implies V^\top(\vec{x}_t - \vec{x}_\lambda^\star) = \left(I - \eta\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)\right)^t V^\top(\vec{x}_0 - \vec{x}_\lambda^\star) \tag{25}$$

$$\implies \vec{z}_t - \vec{z}_\lambda^\star = \left(I - \eta\left(\frac{\Sigma^\top\Sigma}{m} + \lambda I\right)\right)^t (\vec{z}_0 - \vec{z}_\lambda^\star). \tag{26}$$

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Now note that the quantity $I - \eta(\frac{\Sigma^\top \Sigma}{m} + \lambda I)$ is a diagonal matrix. Thus we have

$$\vec{z}_t - \vec{z}_\lambda^\star = \left(I - \eta\left(\frac{\Sigma^\top \Sigma}{m} + \lambda I\right)\right)^t (\vec{z}_0 - \vec{z}_\lambda^\star) \tag{27}$$

$$\implies (\vec{z}_t - \vec{z}_\lambda^\star)_i = \left(I - \eta\left(\frac{\Sigma^\top \Sigma}{m} + \lambda I\right)\right)_i^t (\vec{z}_0 - \vec{z}_\lambda^\star)_i \tag{28}$$

$$\implies (\vec{z}_t)_i - (\vec{z}_\lambda^\star)_i = \left(1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right)^t ((\vec{z}_0)_i - (\vec{z}_\lambda^\star)_i) \tag{29}$$

(d) Show that $\lim_{t\to\infty} \vec{z}_t = \vec{z}_\lambda^\star$ for all initializations $\vec{x}_0 = V\vec{z}_0$ if and only if

$$\max_{i\in\{1,\ldots,n\}} \left|1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right| < 1. \tag{30}$$

Use this to show that GD converges for all initializations $\vec{x}_0$ if and only if

$$\eta \in \left(0, \frac{2m}{\sigma_{\max}\{A\}^2 + \lambda m}\right) \tag{31}$$

where $\sigma_{\max}\{A\} = \sigma_1\{A\}$ is the largest singular value of A.

Solution: We have

$$\lim_{t\to\infty} \vec{z}_t = \vec{z}_\lambda^\star, \qquad \forall \vec{x}_0 \tag{32}$$

$$\iff \lim_{t\to\infty} (\vec{z}_t)_i = (\vec{z}_\lambda^\star)_i, \qquad \forall i \quad \forall \vec{x}_0 \tag{33}$$

$$\iff \lim_{t\to\infty} \left(1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right)^t = 0, \qquad \forall i \tag{34}$$

$$\iff \left|1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right| < 1, \qquad \forall i \tag{35}$$

$$\iff \max_{i\in\{1,\ldots,n\}} \left|1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right| < 1. \tag{36}$$

This proves the first part of the question. The second part of the question follows by noting that

$$\max_{i\in\{1,\ldots,n\}} \left|1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right| < 1 \tag{37}$$

$$\iff \max_{i\in\{1,\ldots,n\}} \left(1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right) < 1 \tag{38}$$

$$\text{and} \quad \min_{i\in\{1,\ldots,n\}} \left(1 - \eta\left(\frac{\sigma_i\{A\}^2}{m} + \lambda\right)\right) > -1 \tag{39}$$

$$\iff 1 - \eta\left(\frac{\sigma_{\min}\{A\}^2}{m} + \lambda\right) < 1 \tag{40}$$

$$\text{and} \quad 1 - \eta\left(\frac{\sigma_{\max}\{A\}^2}{m} + \lambda\right) > -1. \tag{41}$$

Now the first equation is always satisfied for $\eta > 0$ and $\lambda > 0$ because $\frac{\sigma_{\min}\{A\}^2}{m} + \lambda > 0$ so $1 - \eta(\frac{\sigma_{\min}\{A\}^2}{m} + \lambda) < 1$. The second equation is satisfied when $\eta < \frac{2m}{\sigma_{\max}\{A\}^2 + \lambda m}$. Since $\lim_{t\to\infty} \vec{x}_t = \vec{x}_\lambda^\star$ if and only if $\lim_{t\to\infty} \vec{z}_t = \vec{z}_\lambda^\star$, we have that gradient descent converges for all initializations $\vec{x}_0$ if and only if $0 < \eta < \frac{2m}{\sigma_{\max}\{A\}^2 + \lambda m}$.

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(e) **(OPTIONAL)** One way we can derive an "optimal" learning rate $\eta^\star$ is to minimize the largest rate of convergence:

$$\eta^\star \in \operatorname*{argmin}_{\eta\in\mathbb{R}} \max_{i\in\{1,\ldots,n\}} \left|1-\eta\left(\frac{\sigma_i\{A\}^2}{m}+\lambda\right)\right|. \tag{42}$$

One important property of $\eta^\star$ is that it makes the rates of convergence $\left|1-\eta\left(\frac{\sigma_i\{A\}^2}{m}+\lambda\right)\right|$ associated with the largest and smallest singular values of A equal. Use this property to show that

$$\eta^\star = \frac{2m}{\sigma_{\max}\{A\}^2+\sigma_{\min}\{A\}^2+2\lambda m} \tag{43}$$

where $\sigma_{\min}\{A\} = \sigma_n\{A\}$ is the nth largest singular value of A.

NOTE: There are several useful notions of optimal learning rate; this is just one of them.

Solution: We have

$$\left|1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right)\right| = \left|1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)\right| \tag{44}$$

$$1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right) = -\left(1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)\right) \tag{45}$$

$$1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right) = \eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)-1 \tag{46}$$

$$2 = \eta^\star\left(\frac{\sigma_{\max}\{A\}^2+\sigma_{\min}\{A\}^2}{m}+2\lambda\right) \tag{47}$$

$$\eta^\star = \frac{2m}{\sigma_{\max}\{A\}^2+\sigma_{\min}\{A\}^2+2m\lambda}. \tag{48}$$

Here the second equality is the most challenging to derive. It follows from the first inequality by the following reasoning:

- If $1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right)$ and $1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)$ have the same sign, then by the first equality, they must be equal. This means that $\sigma_{\max}\{A\} = \sigma_1\{A\} = \sigma_2\{A\} = \cdots = \sigma_n\{A\} = \sigma_{\min}\{A\}$ and the optimal learning rate $\eta^\star$ sets each rate $1-\eta^\star\left(\frac{\sigma_i\{A\}^2}{m}+\lambda\right)$ to 0 simultaneously, ensuring convergence in one step. If both sides are 0 then the second equality holds (because $0=-0$).
- Otherwise, $1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right)$ and $1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)$ have opposite signs. Since $\sigma_{\max}\{A\} > \sigma_{\min}\{A\}$ (since if they were equal we would be in the first case), we have $1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right) > 1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)$. Thus $1-\eta^\star\left(\frac{\sigma_{\min}\{A\}^2}{m}+\lambda\right)$ must be positive and $1-\eta^\star\left(\frac{\sigma_{\max}\{A\}^2}{m}+\lambda\right)$ must be negative. The absolute value of a negative number is its negative, so the second equality follows directly from the first equality.

(f) The Google Colab notebook, gd_convergence.ipynb, will examine the computational aspects of GD on ridge regression. Implement the GD and stochastic gradient descent (SGD) functions at the top of the notebook, which are marked with `TODO`s.

(g) Click through the notebook and run the sections $n=1$, $n=2$, and $n \gg 2$. Change the values of λ and η and re-run the cells a few times. Write down your observations about how the convergence of GD works under different values of λ and η.

Solution: See the solution Google Colab notebook. We know from previous parts that when $\eta \geq \left(\frac{2m}{\sigma_{\max}\{A\}^2+\lambda m}\right)$, gradient descent does not converge. This is also observed in the notebook that when η is increased by too much, gradient descent does not converge. Moreover, when η is too small, it does not converge within the prescribed number of iterations either.

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(h) In the sections $n = 1$, $n = 2$, and $n \gg 2$, change the calls to GD to instead call SGD. Write down your observations about how the convergence of SGD works under different values of λ and η. Compare the behavior of GD and SGD.

Solution: It is much harder to tune λ and η to make stochastic gradient descent converge. Due to the randomness, the trajectory is much noisier than gradient descent.

(i) You might have noticed that if we think of convergence in the "last iterate" sense, i.e., $\lim_{T\to\infty} \vec{x}_T = \vec{x}^\star_\lambda$, then *SGD rarely converges*. This is because even if we reach the global optimum, the gradient estimate used by SGD is in general nonzero, and so the iterates end up bouncing around near the optimum. Another different, weaker, notion of convergence under which one might show that SGD actually does converge is convergence "in time average", i.e., $\lim_{T\to\infty} \overline{\vec{x}}_T = \vec{x}^\star_\lambda$ where $\overline{\vec{x}}_T \doteq \frac{1}{T}\sum_{t=1}^{T} \vec{x}_t$. Visualize this by adding the argument `time_avg=True` to each plotting function; the plot will now visualize the sequence of $\overline{\vec{x}}_t$. Re-run the notebook. Write down your observations, especially regarding the stability of SGD and convergence in the last-iterate sense versus the time-average sense.

Solution: When η is set too large, the trajectories don't converge no matter when `time_avg=True` or when `time_avg=False`. On the other hand, when η is not set too large, the trajectories under `time_avg=True` don't bounce around that much near the optimum compared with when `time_avg=False`. Instead, the closer $\vec{x}$ is to the optimum, the more stablized it becomes, which shows that it converges in the limit.

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2. Convexity of Rank 1 Matrices

In this problem, we explore the effect of rank constraints on the convexity of matrix sets.

First, consider the set of all 2×2 matrices with diagonal elements $(1, 2)$, which we can write explicitly as

$$\mathcal{S} = \left\{ \begin{bmatrix} 1 & x \\ y & 2 \end{bmatrix} \middle| \, x, y \in \mathbb{R} \right\}. \tag{49}$$

(a) Is set $\mathcal{S}$ convex? If so, provide a proof, and if not, provide a counterexample.

Solution: Set $\mathcal{S}$ is indeed convex. In general, a set S is convex if for all elements $s_1, s_2 \in S$, and $\lambda \in [0, 1]$,

$$\lambda s_1 + (1 - \lambda) s_2 \in S. \tag{50}$$

Considering $s_1 = \begin{bmatrix} 1 & x_1 \\ y_1 & 2 \end{bmatrix}, s_2 = \begin{bmatrix} 1 & x_2 \\ y_2 & 2 \end{bmatrix} \in \mathcal{S}$, we have

$$\begin{aligned} \lambda s_1 + (1 - \lambda) s_2 &= \lambda \begin{bmatrix} 1 & x_1 \\ y_1 & 2 \end{bmatrix} + (1 - \lambda) \begin{bmatrix} 1 & x_2 \\ y_2 & 2 \end{bmatrix} && (51) \\ &= \begin{bmatrix} 1\lambda + 1(1 - \lambda) & \lambda x_1 + (1 - \lambda) x_2 \\ \lambda y_1 + (1 - \lambda) y_2 & 2\lambda + 2(1 - \lambda) \end{bmatrix} && (52) \\ &= \begin{bmatrix} 1 & \lambda x_1 + (1 - \lambda) x_2 \\ \lambda y_1 + (1 - \lambda) y_2 & 2 \end{bmatrix} && (53) \\ &\in \mathcal{S}. && (54) \end{aligned}$$

(b) Suppose we now wish to define $\mathcal{S}_1 \subset \mathcal{S}$, the set of all rank-1 matrices in $\mathcal{S}$. Write out conditions on x and y (i.e. equation constraints that x and y must satisfy) to define $\mathcal{S}_1$ explicitly.

Solution: If a matrix $s = \begin{bmatrix} 1 & x \\ y & 2 \end{bmatrix}$ is an element of $\mathcal{S}_1$, then its columns must be linearly dependent, i.e.,

$$\begin{bmatrix} x \\ 2 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ y \end{bmatrix} \tag{55}$$

for some $\alpha \in \mathbb{R}$. Computing each row equality, we have $x = \alpha$ and $2 = \alpha y \Rightarrow y = \frac{2}{\alpha}$, and thus

$$\begin{aligned} \mathcal{S}_1 &= \left\{ \begin{bmatrix} 1 & x \\ y & 2 \end{bmatrix} \middle| \, x = \alpha, y = \frac{2}{\alpha}, \alpha \in \mathbb{R} \right\} && (56) \\ &= \left\{ \begin{bmatrix} 1 & x \\ y & 2 \end{bmatrix} \middle| \, x, y \in \mathbb{R}, y = \frac{2}{x} \right\} && (57) \end{aligned}$$

or alternatively

$$\mathcal{S}_1 = \left\{ \begin{bmatrix} 1 & \alpha \\ \frac{2}{\alpha} & 2 \end{bmatrix} \middle| \, \alpha \in \mathbb{R} \right\}. \tag{58}$$

Note that this is equivalent to

$$\mathcal{S}_1 = \left\{ \begin{bmatrix} 1 & \frac{1}{\beta} \\ 2\beta & 2 \end{bmatrix} \middle| \, \beta \in \mathbb{R} \right\}, \tag{59}$$

which we can see by defining $\beta \doteq \frac{1}{\alpha}$.

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(c) Is set $\mathcal{S}_1$ convex? If so, provide a proof, and if not, provide a counterexample.

Solution: Set $\mathcal{S}_1$ is **not** convex.

Let $s_1, s_2 \in S_1$ and $s_1 = \begin{bmatrix} 1 & x \\ \frac{2}{x} & 2 \end{bmatrix}, s_2 = \begin{bmatrix} 1 & y \\ \frac{2}{y} & 2 \end{bmatrix}$. Let $0 \leq \lambda \leq 1, s_3(\lambda) = \lambda s_1 + (1-\lambda)s_2 = \begin{bmatrix} 1 & \lambda x + (1-\lambda)y \\ \frac{2\lambda}{x} + \frac{2(1-\lambda)}{y} & 2 \end{bmatrix}$. Since $\exists \lambda . 0 \leq \lambda \leq 1$ such that $(\lambda x + (1-\lambda)y)(\frac{2\lambda}{x} + \frac{2(1-\lambda)}{y}) \neq 2$, S_1 is not convex.

For one possible counterexample, consider $s_1 = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, s_2 = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \in \mathcal{S}_1, \lambda = \frac{1}{2}$. Then

$$\lambda s_1 + (1-\lambda)s_2 = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \left(1 - \frac{1}{2}\right)\begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \tag{60}$$

$$= \begin{bmatrix} 1 & -\frac{1}{2} \\ \frac{1}{2} & 2 \end{bmatrix} \tag{61}$$

$$\notin \mathcal{S}_1. \tag{62}$$

(d) In this class, we will sometimes pose optimization problems in which we optimize over sets of matrices. Since low-dimensional models are often easier to interpret, it would be nice to impose rank constraints on these solution matrices. Suppose we wish to solve the optimization problem

$$\min_{A \in \mathcal{S}_1} \|A\|_F^2 \tag{63}$$

which is equivalent to

$$\min_{A \in \mathcal{S}} \quad \|A\|_F^2 \tag{64}$$

$$\text{s.t.} \quad \operatorname{rank}(A) = 1. \tag{65}$$

Is this optimization problem convex? (That is, are the objective function $\|A\|_F^2$ and feasible set $\mathcal{S}_1$ convex?)

Solution: While optimizing $\|A\|_F^2$ over the set of all $A \in \mathbb{R}^{2\times 2}$ is convex, because $\mathcal{S}_1$ is not convex, this optimization problem is **not** convex because the domain of the function is not convex.

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3. Positive Semi-Definite Cone

Let $A, B \in \mathbb{R}^{n\times n}$ be two symmetric positive semidefinite matrices. Show that $\langle A,\ B\rangle \geq 0$, where $\langle A,\ B\rangle = \operatorname{tr}\left(AB^\top\right)$ is the usual inner product on matrices.

We will see in lecture that the set of symmetric positive semidefinite matrices $\mathbb{S}^n_+$ is a so-called *convex* set within $\mathbb{R}^{n\times n}$. This problem shows that $\mathbb{S}^n_+$ is in fact a geometric object called a *cone* in $\mathbb{R}^{n\times n}$.

HINT: Try an eigendecomposition of $B = \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top$ and use the trace formula of the inner product.

Solution:

Solution 1:

Assume B has the eigendecomposition $B = \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top$, where $\lambda_i \geq 0$ for all $i = 1, \ldots, n$, and $\{\vec{u}_i\}_{i=1}^n$ is an orthonormal set of vectors. Then,

$$\begin{aligned}
\langle A,\ B\rangle &= \left\langle A,\ \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top \right\rangle && (66)\\
&= \operatorname{tr}\left(A \sum_{i=1}^n \lambda_i \vec{u}_i \vec{u}_i^\top\right) && (67)\\
&= \operatorname{tr}\left(\sum_{i=1}^n \lambda_i A\vec{u}_i \vec{u}_i^\top\right) && (68)\\
&= \sum_{i=1}^n \lambda_i \operatorname{tr}\left(A\vec{u}_i \vec{u}_i^\top\right) && (69)\\
&= \sum_{i=1}^n \lambda_i \operatorname{tr}\left(\vec{u}_i^\top A\vec{u}_i\right) && (70)\\
&= \sum_{i=1}^n \lambda_i \underbrace{\vec{u}_i^\top A\vec{u}_i}_{\geq 0} && (71)\\
&\geq 0. && (72)
\end{aligned}$$

Solution 2:

Let $A = C^\top C$ and $B = R^\top R$. Then

$$\begin{aligned}
\langle A,\ B\rangle &= \operatorname{tr}\left(A^\top B\right) && (73)\\
&= \operatorname{tr}\left(C^\top C R^\top R\right) && (74)\\
&= \operatorname{tr}\left(R C^\top C R^\top\right) && (75)\\
&= \operatorname{tr}\left((CR^\top)^\top CR^\top\right) && (76)\\
&= \left\|CR^\top\right\|_F^2 && (77)\\
&\geq 0 && (78)
\end{aligned}$$

where the third equivalence follows from the cyclic property of the trace.

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4. Visualizing the Dual Problem

Use the associated Google Colab notebook; complete the code where designated and answer the questions given in the space provided. (If you prefer, for questions that do not involve writing code, you can write solutions on separate paper or LaTeX PDF, just make sure to correctly mark the relevant pages when uploading to Gradescope.)

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5. Dual of an LP

Consider the general form of a linear program:

$$\min_{\vec{x}} \quad \vec{c}^\top \vec{x} \tag{79}$$

$$\text{s.t.} \quad A\vec{x} = \vec{b} \tag{80}$$

(a) Write the Lagrangian function $\mathcal{L}(\vec{x}, \vec{\mu})$.

Solution:

$$\mathcal{L}(\vec{x}, \vec{\mu}) = \vec{c}^\top \vec{x} + \vec{\mu}^\top (A\vec{x} - \vec{b}) \tag{81}$$

(b) Write the Lagrangian dual function $g(\vec{\mu})$.

Solution:

$$g(\mu) = \inf_{\vec{x}} \ \mathcal{L}(\vec{x}, \vec{\mu}) = \begin{cases} -\vec{b}^\top \vec{\mu} & \text{if } \vec{c} + A^\top \vec{\mu} = \vec{0} \\ -\infty & \text{otherwise} \end{cases} \tag{82}$$

(c) Write the dual problem.

Solution: Substituting, we get

$$d^\star = \max_{\vec{\mu}} \quad -\vec{b}^\top \vec{\mu} \tag{83}$$

$$\text{s.t.} \quad A^\top \vec{\mu} + \vec{c} = \vec{0} \tag{84}$$

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6. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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